

Program - 3

Code a Dockerized Python Flask or Node.js Application

Project Structure:-

Program-3/

-->Dockerfile

-->requirements.txt

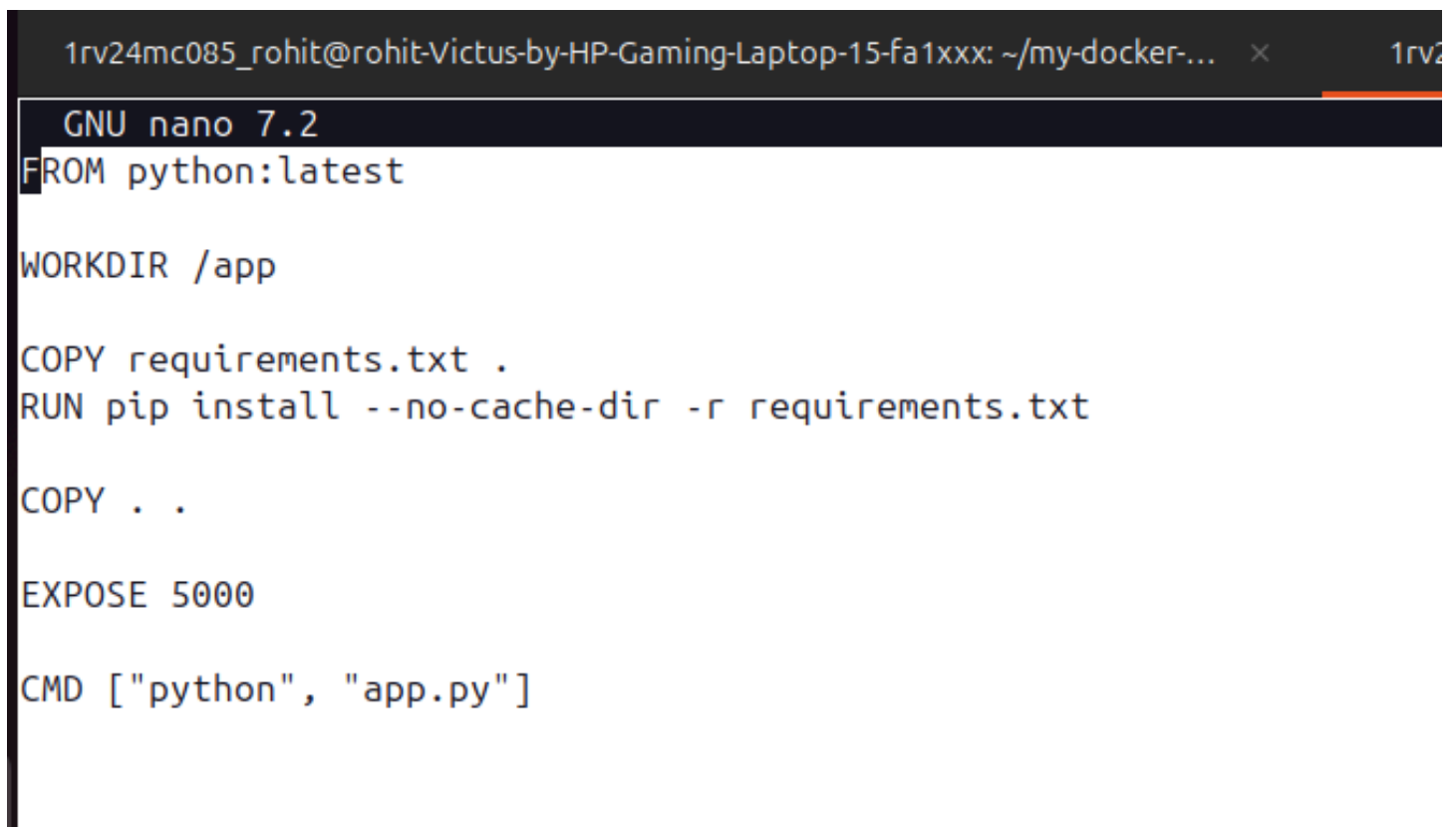
-->app.py

Step 1:Create a Dockerfile and add the content

mkdir Program-3

cd Program-3

nano Dockerfile

A screenshot of a terminal window with a dark background. The window title bar shows the user '1rv24mc085_rohit' on a machine named 'rohit-Victus-by-HP-Gaming-Laptop-15-fa1xxx' in the directory '~/my-docker-...'. The terminal shows the 'GNU nano 7.2' prompt. The content of the Dockerfile being edited is as follows:

```
FROM python:latest

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python", "app.py"]
```

Step 2:Make a file app.py

nano app.py

GNU nano 7.2

```
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from Simple Flask Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Step 3: Create a file requirements.txt

nano requirements.txt

GNU nano 7.2

Flask==2.3.3

Step 4: Execute docker build command

`sudo docker build -t program-3 .`

```
Rohit@JD:~/Program-3$ docker build -t program-3 .
[+] Building 1.6s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 197B
=> [internal] load metadata for docker.io/library/python:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 93B
=> [1/5] FROM docker.io/library/python:latest@sha256:1ad1a43b5e2478e62056bbc28028afd858185d73bf4d6a439cbb058b6800a96d
=> => resolve docker.io/library/python:latest@sha256:1ad1a43b5e2478e62056bbc28028afd858185d73bf4d6a439cbb058b6800a96d
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:bd4c9ebbfbc89366658992c3059476cf1c6fcc53c53077c9f49fc40016df346
=> => naming to docker.io/library/program-3
```

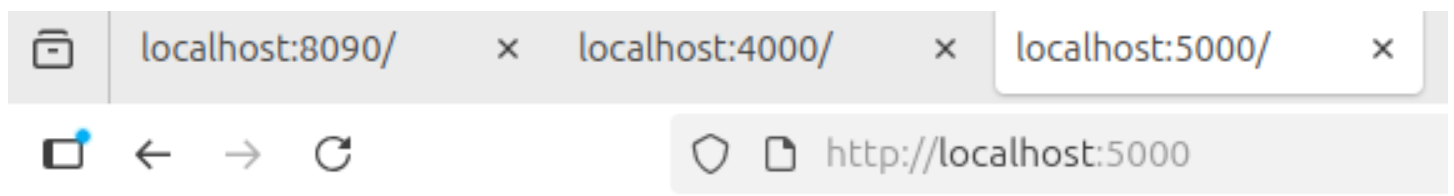
Step 5: Execute docker run command

`sudo docker run -d -p 5000:5000 program-3`

```
Rohit@JD:~/Program-3$ docker run -d -p 5000:5000 program-3
489d4916cf60c65e2091c068591089d53d3b1c4bd185aaa1727616624fb54ce9
```

Step 6: Test the container in the browser through localhost

`http://localhost:5000`



Hello from Simple Flask Docker!

